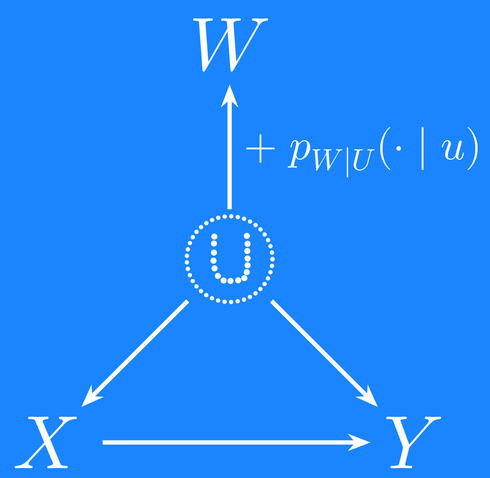
We identify causal effects

under unobserved confounding with a known *error mechanism*.



Single Proxy Identifiability of Causal Effects

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structural causal model

$$
\begin{aligned}
U &\leftarrow N_U \\
W &\leftarrow AU + E \\
X &\leftarrow f_X(U, N_X) \\
Y &\leftarrow f_Y(U, X, N_Y)
\end{aligned}
$$

- The proxy $W \in \mathcal{W} \subseteq \mathbb{R}^d$ has at least the dimension of the confounder $U \in \mathcal{U} \subseteq \mathbb{R}^k$, that is, $d \geq k$.
- $U \perp\!\!\!\perp E$ and $A \in \mathbb{R}^{d \times k}$ has full column rank.
- The Fourier transform of the density of E has no zeros.

causal function

Let M_0 be the true SCM. We define the causal function $\theta^{M_0} : \mathcal{X} \to \mathcal{Y}$ for all $x \in \mathcal{X}$ as

$$\theta^{M_0}(x) := \mathbb{E}_Y^{M_0; do(X:=x)}[Y] = \int y \int p^{M_0}_{Y|U,X}(y \mid u, x)\, p^{M_0}_U(u)\, \mu(du, dy).$$

characteristic error mechanism

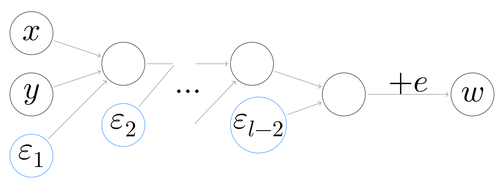
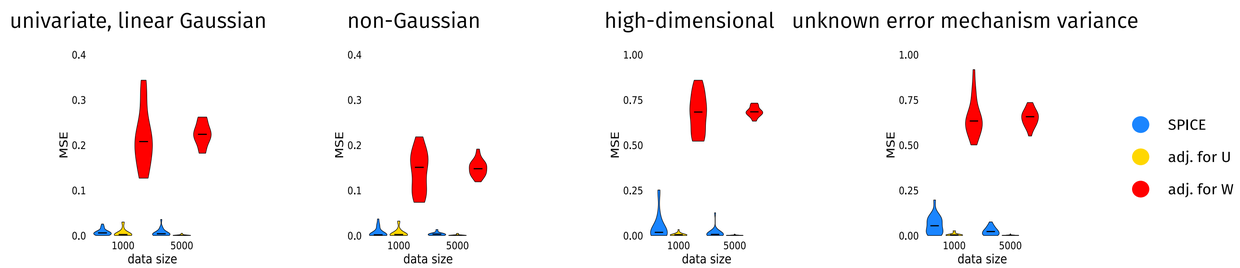
We show that the error mechanism $\{p^{M_0}_{W|U}(\cdot \mid u) \mid u \in \mathcal{U}\}$ is L_∞-characteristic, that is, for all $\delta \in L_\infty(\mathcal{U})$

$$\int_{\mathcal{U}} p^{M_0}_{W|U}(\cdot \mid u)\delta(u)\, \mu(du) \equiv 0 \quad \Longrightarrow \quad \delta \equiv 0 \quad \text{a.e.}$$

identifiability of the causal function

If the error mechanism is L_∞-characteristic and if for all SCMs, which are observationally equivalent to M_0, it holds that $p^{M}_{U|X,Y}(\cdot \mid x, y) \in L_\infty$ $\Longrightarrow$ θ^{M_0} is identifiable.

learn the causal function



1. Learn $P^{M_0}_{\widetilde{A}U|(X,Y)}$ by minimizing the Energy loss as in Engression [Shen and Meinshausen, 2023].
2. Adjust for $\widetilde{A}u$ with your favourite causal function estimator, e.g. EconML [Battocchi et al., 2019].